

APOLLO INSTITUTE OF COMPUTER APPLICATION

BACHELOR OF COMPUTER APPLICATION

SEMESTER 03

Design Thinking and Innovation (BC03001071)

Lecture Notes

Unit 1: Introduction to Design Thinking (5 Hours)

Lecture 1: Introduction to Design Thinking

Learning Outcome

After completing this lecture, students will be able to:

- Understand the meaning of Design Thinking.
- Explain why Design Thinking is important.
- Identify the objectives and characteristics of Design Thinking.

1. What is Design Thinking?

Design Thinking is a **human-centered approach** to solving problems. Instead of focusing only on technology or business goals, it begins by understanding the needs, experiences, and challenges of users. The goal is to create innovative, practical, and user-friendly solutions.

Unlike traditional problem-solving methods, Design Thinking encourages creativity, teamwork, experimentation, and continuous improvement.

Definition

Design Thinking is a creative and systematic process that focuses on understanding users, identifying problems, generating ideas, developing prototypes, and testing solutions to create innovative products and services.

Why Do We Need Design Thinking?

Today's world is changing rapidly because of:

- Technological advancements

- Changing customer expectations
- Global competition
- Digital transformation
- Increasing complexity of problems

Traditional methods often focus only on solving technical issues, while Design Thinking focuses on solving **human problems**.

Benefits

- Understands customer needs.
 - Encourages innovation.
 - Reduces project risks.
 - Saves time and cost.
 - Improves customer satisfaction.
 - Promotes teamwork.
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Objectives of Design Thinking

- Understand users deeply.
 - Solve real-world problems.
 - Generate innovative ideas.
 - Improve products and services.
 - Encourage collaboration.
 - Create practical solutions.
-

Characteristics of Design Thinking

- Human-centered
 - Creative
 - Collaborative
 - Iterative
 - Experimental
 - Solution-oriented
 - User-focused
-

Real-Life Example

A hospital receives complaints about long waiting times.

Using Design Thinking:

- Observe patients.
- Interview doctors and staff.
- Identify bottlenecks.
- Develop an online appointment system.
- Test it with patients.
- Improve based on feedback.

Result: Reduced waiting time and increased patient satisfaction.

Summary

Design Thinking helps organizations understand users, solve problems creatively, and develop better products and services.

Lecture 2: Origin and Evolution of Design Thinking

Learning Outcome

Students will be able to:

- Explain the history of Design Thinking.
 - Identify important contributors.
 - Understand how Design Thinking evolved over time.
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Origin of Design Thinking

The roots of Design Thinking can be traced back to **industrial design and engineering** during the mid-20th century. Initially, designers focused mainly on making products attractive. Later, they realized that products should also solve users' real problems.

This shift from **product-centered design** to **user-centered design** gave birth to Design Thinking.

Evolution Timeline

Period	Development
1950s	Creative problem-solving methods emerged.

Period	Development
1960s	Focus shifted toward understanding users.
1969	Herbert Simon introduced design as a structured problem-solving process.
1980s	Peter Rowe popularized the term "Design Thinking."
1990s	IDEO adopted Design Thinking for business innovation.
2005	Stanford d.school formalized the five-step Design Thinking model.
Today	Used in healthcare, education, business, government, IT, and startups.

Major Contributors

Herbert Simon

- Considered one of the pioneers of Design Thinking.
- Introduced the concept of systematic problem-solving.
- Emphasized decision-making based on human needs.

Robert McKim

- Promoted creative and visual thinking.
- Encouraged designers to use sketches and models.

Peter Rowe

- Popularized the term **Design Thinking**.
- Explained how designers approach complex problems.

IDEO

- A global design consulting company.
- Applied Design Thinking to create innovative products and services.

Stanford d.school

- Developed the popular five-stage Design Thinking process.
 - Helped spread Design Thinking worldwide.
-

Modern Applications

Today, Design Thinking is used in:

- Software development
- Healthcare
- Banking
- Education
- Government services

- Mobile app development
 - Artificial Intelligence
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Summary

Design Thinking evolved from product design into a comprehensive approach for solving complex human-centered problems.

Lecture 3: Attributes of Design Thinking

Learning Outcome

Students will be able to identify and explain the major attributes of Design Thinking.

What are Attributes?

Attributes are the important qualities or characteristics that make Design Thinking effective.

Introduction

Attributes of Design Thinking are the key characteristics or qualities that make Design Thinking an effective approach to solving problems. These attributes help designers and organizations create innovative, practical, and user-friendly solutions. They ensure that the final product or service not only solves the problem but also meets the needs and expectations of users.

1. Human-Centered

Meaning:

Human-Centered Design means that the entire design process focuses on the people who will use the product or service. Instead of starting with technology or business goals, designers first understand the users' needs, behaviors, expectations, and challenges.

The objective is to create solutions that are useful, easy to use, and provide a positive user experience.

Key Points

- Focuses on users rather than technology.
- Identifies real user problems.
- Improves customer satisfaction.

- Makes products more accessible and user-friendly.

Example

While designing a **mobile banking app for elderly users**, developers use large buttons, simple menus, high-contrast colors, and voice guidance so that senior citizens can use the app comfortably.

2. Empathy

Meaning:

Empathy is the ability to understand and share the feelings, experiences, and perspectives of users. It is one of the most important attributes of Design Thinking because it helps designers identify problems from the users' point of view.

Designers gather information through interviews, observations, surveys, and discussions.

Key Points

- Understands users' emotions and needs.
- Helps identify hidden problems.
- Builds trust with users.
- Leads to better and more relevant solutions.

Example

Before developing an **online learning platform**, designers interview students and teachers to understand issues such as poor internet connectivity, confusing interfaces, and difficulty accessing study materials.

3. Creativity

Meaning:

Creativity is the ability to generate new, original, and useful ideas. Design Thinking encourages teams to think beyond traditional solutions and explore multiple possibilities before selecting the best one.

There is no restriction on ideas during brainstorming; even unusual ideas are welcomed.

Key Points

- Encourages innovative thinking.
- Generates multiple possible solutions.
- Avoids limiting ideas too early.
- Promotes continuous innovation.

Example

A bicycle manufacturer designs a **foldable bicycle** for people living in crowded cities, making it easier to carry and store in small apartments or public transport.

4. Collaboration

Meaning:

Collaboration means working together with people from different backgrounds, skills, and expertise. Since every person has a different perspective, teamwork often results in more creative and practical solutions.

Design Thinking encourages open communication and sharing of ideas.

Key Points

- Promotes teamwork.
- Combines diverse knowledge and skills.
- Improves decision-making.
- Produces better solutions.

Example

To develop a **healthcare application**, doctors provide medical knowledge, software developers build the application, designers create an easy-to-use interface, and patients provide feedback about usability.

5. Experimentation

Meaning:

Experimentation involves trying different ideas by creating simple prototypes and testing them before developing the final product. This helps identify problems early and reduces the risk of failure.

Experimentation encourages learning through practical experience.

Key Points

- Tests different ideas.
- Identifies mistakes early.
- Reduces development cost.
- Improves product quality.

Example

A company creates **three different website layouts** and asks users to test them. Based on user preferences and feedback, the best design is selected.

6. Iteration

Meaning:

Iteration means continuously improving a solution by testing it, collecting feedback, and making necessary changes. Design Thinking does not assume that the first solution is perfect. Instead, products are refined until they meet user expectations.

Design Cycle

Prototype

↓

Test

↓

Collect Feedback

↓

Improve

↓

Retest

Key Points

- Continuous improvement.
- Uses user feedback.
- Corrects errors quickly.
- Produces high-quality solutions.

Example

A mobile application releases Version 1.0, collects customer reviews, fixes problems, adds new features, and launches Version 2.0.

7. Optimism

Meaning:

Optimism is the belief that every problem has a solution. Design Thinking encourages teams to remain positive even when facing difficult challenges or unsuccessful attempts.

A positive mindset motivates teams to continue experimenting until they discover an effective solution.

Key Points

- Encourages confidence.

- Motivates continuous improvement.
- Supports innovation.
- Accepts challenges positively.

Example

A startup developing a water purification system continues improving its prototype even after several failed tests because the team believes they can eventually create an affordable solution.

8. User Feedback

Meaning:

User feedback is the information and suggestions collected from users after they interact with a product or prototype. Feedback helps designers understand what works well and what needs improvement.

Continuous feedback ensures that the final product truly meets user expectations.

Key Points

- Helps improve products.
- Identifies user satisfaction.
- Reduces design errors.
- Encourages continuous improvement.

Example

A food delivery app introduces a new interface and asks customers to rate their experience. Based on the feedback, the navigation and payment process are simplified.

9. Practicality

Meaning:

Practicality means that a solution should be realistic, affordable, and easy to implement. An innovative idea is valuable only if it can be developed within available resources and effectively solve the intended problem.

Key Points

- Solutions should be feasible.
- Considers budget and available technology.
- Easy to implement and maintain.
- Solves real-world problems.

Example

Instead of building an expensive smart irrigation system, engineers develop a **low-cost automatic watering system** that small farmers can easily afford and maintain.

10. Innovation

Meaning:

Innovation means creating new or improved products, services, or processes that provide value to users. Innovation is not always about inventing something completely new; it can also involve improving existing solutions.

Key Points

- Creates value for users.
- Improves existing products.
- Encourages competitive advantage.
- Solves problems in better ways.

Example

The introduction of **UPI (Unified Payments Interface)** transformed digital payments in India by making money transfers instant, secure, and convenient using smartphones.

Summary Table

Attribute	Meaning	Example
Human-Centered	Focuses on users' needs and experiences.	Mobile app with large buttons for elderly users.
Empathy	Understands users' feelings and challenges.	Interviewing students before designing an e-learning platform.
Creativity	Generates multiple innovative ideas.	Foldable bicycle for city commuters.
Collaboration	Teamwork among people with different expertise.	Doctors and software developers building a healthcare app.
Experimentation	Tests ideas using prototypes.	Comparing different website designs before launch.
Iteration	Continuously improves the solution using feedback.	Updating a mobile app after customer reviews.
Optimism	Believes every problem has a solution.	Improving a water purification system after failed attempts.
User Feedback	Learns from users to improve products.	Collecting customer ratings for a food delivery app.
Practicality	Develops feasible and affordable	Low-cost irrigation system for farmers.

Attribute	Meaning	Example
Innovation	solutions. Creates valuable new or improved solutions.	UPI-based digital payment system.

Lecture 4: Principles of Design Thinking

Learning Outcome

Students will understand the core principles that guide the Design Thinking process.

What are Principles of Design Thinking?

Principles of Design Thinking are the fundamental guidelines or rules that help designers and teams solve problems effectively. These principles ensure that solutions are user-focused, creative, practical, and continuously improved through feedback and testing.

By following these principles, organizations can develop products and services that not only solve problems but also provide a better experience for users.

1. Empathy

Meaning

Empathy is the ability to understand the feelings, needs, experiences, and challenges of users before trying to solve their problems. It helps designers see the problem from the user's perspective instead of making assumptions.

Empathy is the foundation of Design Thinking because a solution can only be effective if it addresses the real needs of users.

Key Points

- Understand users before designing solutions.
- Observe users in real-life situations.
- Conduct interviews, surveys, and discussions.
- Identify users' pain points and expectations.

Example

Before redesigning hospital services, the design team talks to patients, doctors, and nurses to understand issues such as long waiting times, confusing registration procedures, and lack of proper guidance.

2. User-Centered Design

Meaning

User-Centered Design means that every design decision should focus on the users rather than the preferences of designers or developers. The product should be simple, useful, accessible, and easy to use.

A successful product is one that satisfies users' needs and improves their experience.

Key Points

- Focus on users' requirements.
- Design products that are easy to use.
- Improve accessibility and usability.
- Increase customer satisfaction.

Example

A mobile banking app provides a simple interface, large icons, and easy navigation so that users of all age groups can use it without difficulty.

3. Collaboration

Meaning

Collaboration means working together with people from different backgrounds and areas of expertise. Every team member contributes unique ideas, leading to more creative and practical solutions.

Design Thinking encourages open communication, teamwork, and knowledge sharing.

Key Points

- Encourages teamwork.
- Combines different perspectives.
- Improves creativity and innovation.
- Produces better decisions.

Example

While developing a healthcare application, doctors provide medical knowledge, software developers build the application, UI/UX designers improve usability, and patients give feedback on the overall experience.

4. Experimentation

Meaning

Experimentation involves testing different ideas before selecting the final solution. Designers create simple prototypes to check whether an idea works in practice.

Instead of assuming a solution is correct, teams validate their ideas through testing.

Key Points

- Develop prototypes.
- Test multiple ideas.
- Learn from practical results.
- Reduce development risks.

Example

An e-commerce company creates different versions of its checkout page and asks users to test them. The version that provides the best user experience is selected.

5. Bias Toward Action

Meaning

Bias Toward Action means taking action quickly instead of spending too much time planning or discussing. Design Thinking encourages teams to build simple prototypes, test ideas early, and learn through experience.

Quick action helps identify problems faster and saves time.

Key Points

- Start with simple solutions.
- Avoid unnecessary delays.
- Learn by doing.
- Improve through testing.

Example

Instead of spending months discussing a new mobile app design, the team develops a basic version within a week and collects user feedback for further improvement.

6. Iterative Learning

Meaning

Iterative Learning is the process of continuously improving a solution through repeated testing and feedback. Designers understand that the first version is rarely perfect, so they keep refining the product until it meets user expectations.

Design Cycle

Prototype



Test



Collect Feedback



Improve



Retest

Key Points

- Continuous improvement.
- Learn from user feedback.
- Correct mistakes early.
- Increase product quality.

Example

A food delivery application regularly updates its features based on customer reviews, adding improvements such as faster search, better tracking, and easier payment options.

7. Show, Don't Tell

Meaning

Instead of explaining ideas only with words, Design Thinking encourages teams to present ideas visually using sketches, diagrams, wireframes, prototypes, or demonstrations.

Visual communication makes ideas easier to understand and evaluate.

Key Points

- Use visual representations.
- Improve communication.
- Make feedback easier.
- Help users understand ideas quickly.

Example

An architect presents a 3D model of a new building rather than only describing the design verbally, allowing clients to visualize the final structure.

8. Learn from Failure

Meaning

Design Thinking considers failure as an opportunity to learn rather than a reason to stop. Every unsuccessful attempt provides valuable information that helps improve the next version of the solution.

Learning from mistakes leads to stronger and more effective designs.

Key Points

- Accept mistakes positively.
- Identify weaknesses.
- Improve future solutions.
- Encourage innovation.

Example

A startup launches a mobile application that receives poor user ratings. Instead of abandoning the project, the team studies customer feedback, fixes the issues, and releases an improved version.

9. Holistic Thinking

Meaning

Holistic Thinking means considering the entire user experience instead of focusing on only one part of the product or service. Every interaction that a user has with the system should work smoothly.

Designers consider the complete journey from beginning to end.

Key Points

- Focus on the complete user journey.

- Connect all stages of the experience.
- Improve overall customer satisfaction.
- Identify problems across the entire system.

Example

When designing a food delivery app, the team considers every stage of the customer experience:

- Ordering food
- Online payment
- Restaurant confirmation
- Live delivery tracking
- Customer support
- Feedback and ratings

If any one of these stages is poor, the overall user experience is affected.

Summary Table

Principle	Meaning	Example
Empathy	Understand users before solving problems.	Interviewing patients before redesigning hospital services.
User-Centered Design	Focus on users' needs and experiences.	Simple and easy-to-use banking app.
Collaboration	Work together with experts from different fields.	Doctors, designers, and developers creating a healthcare app.
Experimentation	Test ideas through prototypes.	Trying different website layouts before launch.
Bias Toward Action	Build and test quickly instead of waiting for perfection.	Launching a basic app version for early feedback.
Iterative Learning	Continuously improve using testing and feedback.	Updating an app after customer reviews.
Show, Don't Tell	Use sketches and prototypes to explain ideas.	Showing a 3D building model to clients.
Learn from Failure	Treat failures as learning opportunities.	Improving an app after receiving poor ratings.
Holistic Thinking	Consider the complete user experience.	Designing every stage of a food delivery service.

Advantages of Following Design Thinking Principles

- Helps understand users better.
 - Encourages innovation and creativity.
 - Improves teamwork and collaboration.
 - Reduces development risks through testing.
 - Produces user-friendly products and services.
 - Saves time and development cost.
 - Increases customer satisfaction.
 - Supports continuous improvement.
-

Conclusion

The principles of Design Thinking provide a structured approach to solving real-world problems. By practicing **empathy, user-centered design, collaboration, experimentation, bias toward action, iterative learning, visual communication, learning from failure, and holistic thinking**, organizations can create innovative, practical, and user-friendly solutions.

Lecture 5: Models of Design Thinking

Learning Outcome

Students will compare different Design Thinking models and understand where they are used.

Why Different Models?

Different organizations developed their own Design Thinking frameworks, but all models share a common objective: understanding users and solving problems creatively.

Stanford d.school Model (Most Popular)

The **Stanford d.school Model** is the most popular Design Thinking model. It was developed by the **Stanford d.school (Hasso Plattner Institute of Design)**. This model provides a **human-centered approach** to solving problems by understanding users, generating ideas, creating prototypes, and testing solutions.

It consists of **five stages**, and the process is **iterative**, meaning teams can return to any previous stage to improve the solution based on feedback.

Steps

Empathize



Define



Ideate



Prototype



Test

1. Empathize

Meaning:

The first step is to understand the users, their needs, feelings, and challenges. Designers collect information by interacting with users.

Activities:

- Interviews
- Observation
- Surveys

Example:

Interviewing students to understand the problems they face while using an online learning platform.

2. Define

Meaning:

After understanding the users, the team analyzes the information and clearly identifies the real problem that needs to be solved.

Activities:

- Analyze user feedback
- Identify the main problem
- Write a problem statement

Example:

Students find it difficult to locate study materials because the website navigation is confusing.

3. Ideate

Meaning:

In this stage, the team generates many creative ideas to solve the identified problem. Brainstorming is commonly used to encourage innovative thinking.

Activities:

- Brainstorming
- Mind Mapping
- Group Discussion

Example:

The team suggests adding a search bar, subject-wise categories, and AI-based recommendations to improve the learning platform.

4. Prototype

Meaning:

A prototype is a simple or preliminary version of the solution. It helps test ideas before developing the final product.

Activities:

- Paper sketches
- Wireframes
- Basic application model

Example:

Creating a clickable prototype of the redesigned online learning portal.

5. Test

Meaning:

The prototype is tested by real users to collect feedback. Based on their suggestions, the solution is improved.

Activities:

- User testing
- Collect feedback
- Improve the design

Example:

Students test the new portal and suggest adding semester-wise filters, which are included in the final version.

Advantages of the Stanford d.school Model

- Focuses on users' needs.
 - Encourages creativity and innovation.
 - Promotes teamwork.
 - Reduces development risks through testing.
 - Produces user-friendly solutions.
-

Summary Table

Stage	Purpose
Empathize	Understand users and their needs
Define	Identify the real problem
Ideate	Generate creative ideas
Prototype	Build a simple model
Test	Evaluate and improve the solution

2. IDEO Model

The **IDEO Model** is a popular Design Thinking framework developed by **IDEO**, a global design and innovation company. It focuses on **understanding users, generating creative ideas, testing solutions, and continuously improving them**. This model emphasizes **continuous innovation, collaboration, and learning** throughout the design process.

Steps

Discover

↓

Interpret

↓

Ideate

↓

Experiment

↓

1. Discover

Meaning:

In this stage, designers gather information about users, their needs, and the problems they face. This helps the team understand the real situation.

Activities:

- Observe users
- Conduct interviews
- Collect information

Example:

Interviewing customers to understand why they are dissatisfied with an online shopping app.

2. Interpret

Meaning:

The collected information is analyzed to identify the key problems and insights. The team defines what users really need.

Activities:

- Analyze data
- Identify patterns
- Define the problem

Example:

Finding that customers leave the app because the checkout process is too complicated.

3. Ideate

Meaning:

The team generates many creative ideas to solve the identified problem. Brainstorming and group discussions are commonly used.

Activities:

- Brainstorm ideas
- Discuss possible solutions
- Select the best ideas

Example:

Suggesting one-click checkout, digital wallets, and simplified payment options.

4. Experiment

Meaning:

The best ideas are converted into simple prototypes and tested with users. Feedback is collected to identify improvements.

Activities:

- Build prototypes
- Test with users
- Collect feedback

Example:

Creating a prototype of the new checkout page and asking customers to use it.

5. Evolve

Meaning:

Based on user feedback, the solution is improved continuously. The product keeps evolving to meet changing user needs.

Activities:

- Improve the design
- Add new features
- Continue testing

Example:

Updating the shopping app regularly by adding faster payment methods and improving user experience.

Advantages of the IDEO Model

- Focuses on users' needs.
 - Encourages creativity and innovation.
 - Supports continuous learning and improvement.
 - Promotes teamwork and collaboration.
 - Helps develop user-friendly products and services.
-

Summary Table

Step	Purpose
Discover	Understand users and collect information
Interpret	Analyze information and define the problem

Step	Purpose
Ideate	Generate creative solutions
Experiment	Test ideas using prototypes
Evolve	Continuously improve the solution

3. Double Diamond Model

The **Double Diamond Model** was developed by the **Design Council (UK)**. It is a simple and effective Design Thinking model that helps teams solve problems by first **understanding the problem** and then **developing the best solution**.

The model is called **Double Diamond** because it has **two diamond-shaped phases**:

- The **first diamond** focuses on exploring and defining the problem.
- The **second diamond** focuses on developing and delivering the solution.

Steps

Discover

↓

Define

↓

Develop

↓

Deliver

1. Discover

Meaning:

In this stage, the team gathers information to understand the problem from different perspectives. They conduct research, observe users, and collect data to identify users' needs.

Activities:

- User research
- Observation
- Interviews
- Surveys

Example:

A company interviews customers to understand why they are not satisfied with its online shopping website.

2. Define

Meaning:

The information collected during the Discover stage is analyzed to identify the main problem. A clear problem statement is prepared to guide the design process.

Activities:

- Analyze research findings
- Identify key issues
- Define the problem

Example:

The team finds that customers leave the website because the payment process is slow and confusing.

3. Develop

Meaning:

The team generates different ideas and creates possible solutions. These ideas are discussed, improved, and tested to find the most suitable one.

Activities:

- Brainstorm ideas
- Create prototypes
- Test different solutions

Example:

The team designs multiple payment options and a simplified checkout process for users.

4. Deliver

Meaning:

The best solution is finalized, implemented, and launched for users. After implementation, feedback is collected for future improvements.

Activities:

- Final testing
- Launch the solution
- Monitor performance
- Collect user feedback

Example:

The company launches the new checkout system and monitors customer satisfaction after implementation.

Advantages of the Double Diamond Model

- Clearly separates problem analysis and solution development.
 - Encourages thorough research before designing solutions.
 - Supports creativity and innovation.
 - Reduces the chances of solving the wrong problem.
 - Produces practical and user-focused solutions.
-

Summary Table

Step	Purpose
Discover	Research and understand the problem
Define	Clearly identify the real challenge
Develop	Generate and refine possible solutions
Deliver	Implement and evaluate the best solution
•	

4. IBM Design Thinking Model

The **IBM Design Thinking Model** is a Design Thinking framework developed by **IBM**. It is mainly used in **software development, digital products, cloud computing, and business applications**. The model focuses on **understanding user needs, teamwork, continuous feedback, and rapid improvement**.

Unlike other Design Thinking models, IBM emphasizes **clear project goals, regular team reviews, and continuous involvement of real users** throughout the development process.

Key Elements of IBM Design Thinking

Hills

↓

Playbacks

↓

Sponsor Users

1. Hills

Meaning:

A **Hill** is a clear statement of the project's goal from the user's perspective. It describes **what users should be able to achieve** after the project is completed.

Instead of focusing on technical features, Hills focus on the value delivered to users.

Activities:

- Define project objectives.
- Set user-focused goals.
- Prioritize important features.

Example:

"Our goal is to enable customers to complete an online payment in less than one minute."

2. Playbacks

Meaning:

Playbacks are regular review meetings where the project team demonstrates its progress and collects feedback from team members and stakeholders.

These reviews help identify problems early and ensure the project stays on the right track.

Activities:

- Review project progress.
- Demonstrate prototypes.
- Discuss improvements.
- Collect feedback.

Example:

Every week, the software development team presents the latest version of the application and receives suggestions from managers and designers.

3. Sponsor Users

Meaning:

Sponsor Users are real users who actively participate throughout the project. They provide continuous feedback to ensure the product meets actual user needs.

Their involvement helps create user-friendly and practical solutions.

Activities:

- Involve real users.

- Test prototypes.
- Collect user feedback.
- Improve the product.

Example:

A bank invites selected customers to test its new mobile banking application and provide suggestions before the official launch.

Advantages of the IBM Design Thinking Model

- Focuses on real user needs.
- Encourages teamwork and collaboration.
- Provides continuous feedback throughout the project.
- Identifies problems early through regular reviews.
- Helps develop high-quality software and digital products.

Summary Table

Element	Purpose
Hills	Define clear, user-focused project goals
Playbacks	Review progress and collect regular feedback
Sponsor Users	Involve real users throughout the project

Comparison of Models

Model	Steps	Best Used For
Stanford d.school	Empathize → Define → Ideate → Prototype → Test	Education, Product Design
IDEO	Discover → Interpret → Ideate → Experiment → Evolve	Innovation & Services
Double Diamond	Discover → Define → Develop → Deliver	Business & Government
IBM	Hills → Playbacks → Sponsor Users	Software & IT Projects

Case Study: Swiggy – Applying Design Thinking

Introduction

Swiggy is one of India's leading online food delivery platforms. As the number of users increased, many customers complained about **slow delivery and the lack of transparency in tracking their**

orders. Customers wanted to know **where their order was, who was delivering it, and how long it would take to arrive.**

To solve these problems, Swiggy applied the **Stanford d.school Design Thinking Model.**

Problem

Customers wanted a **faster, more reliable, and transparent food delivery experience.** They were frustrated because they could not track their orders in real time and were uncertain about delivery times.

Applying the Design Thinking Process

1. Empathize

Swiggy first tried to understand the needs and problems of its users.

Activities:

- Interviewed customers.
- Talked to delivery partners.
- Collected customer feedback.
- Observed the food delivery process.

Finding:

Customers wanted real-time updates and accurate delivery information.

2. Define

After analyzing the feedback, Swiggy identified the main problem.

Problem Statement:

Customers cannot track their food orders in real time, leading to uncertainty and dissatisfaction.

3. Ideate

The Swiggy team brainstormed different solutions to improve the delivery experience.

Ideas Generated:

- Live GPS tracking.
- Estimated delivery time (ETA).
- Push notifications for every order update.
- Live location of the delivery partner.

The team selected the most practical and user-friendly ideas.

4. Prototype

Swiggy developed a **basic version** of the tracking system.

Prototype Included:

- Live GPS tracking.
- Order status updates.
- Estimated delivery time.
- Delivery partner details.

The prototype was tested before launching it to all users.

5. Test

The new tracking feature was tested with real customers.

Activities:

- Collected user feedback.
 - Measured customer satisfaction.
 - Fixed bugs and improved the tracking system.
 - Added more accurate delivery time predictions.
-

Result

After implementing the new tracking system:

- Customers could track their orders in real time.
 - Delivery status became more transparent.
 - Customer trust and satisfaction increased.
 - Complaints about delivery uncertainty reduced.
 - Swiggy improved its overall user experience.
-

Design Thinking Flow

Empathize

Interview customers and delivery partners

↓

Define

Identify lack of real-time order tracking

↓

Ideate

Brainstorm GPS tracking and delivery notifications

↓

Prototype

Develop a basic live tracking feature

↓

Test

Collect feedback and improve the system

Key Learning

This case study shows that **Design Thinking helps organizations understand users' problems, develop innovative solutions, and continuously improve them through feedback.** By focusing on customer needs, Swiggy was able to provide a **faster, more transparent, and user-friendly food delivery experience.**